

Incident handler's journal

Instructions

As you continue through this course, you may use this template to record your findings after completing an activity or to take notes on what you've learned about a specific tool or concept. You can also use this journal as a way to log the key takeaways about the different cybersecurity tools or concepts you encounter in this course.

Date: October 6th, 2026	**Entry:** 1
Description	Documenting a cybersecurity incident The incident unfolded in two distinct stages: 1. **Detection and Analysis:** The scenario describes how the organization initially identified the ransomware attack. During the analysis phase, the team reached out to external specialists for technical support. 2. **Containment, Eradication, and Recovery:** The scenario outlines the immediate measures implemented to limit the damage, such as powering down affected computer systems. Lacking the internal capacity to handle remediation and recovery independently, the organization engaged outside partners for support.
Tool(s) used	None.
The 5 W's	Capture the 5 W's of an incident. • **Who** caused the incident? An organized group of unethical hackers • **What** happened? A ransomware security incident • **When** did the incident occur? Tuesday at 9:00 a.m. • **Where** did the incident happen? At a U.S. health care company

	• **Why** did the incident happen? The incident happened because an organized group of unethical hackers accessed the company file system by using targeted phishing emails, which were sent to several employees of the company. The phishing emails contained a malicious attachment that installed malware on an employee's computer once it was downloaded. Then, they encrypted critical files and used ransomware tools to request a large amount of money to decrypt the files.
Additional notes	1. How could the health care company prevent an incident like this from occurring again? 2. Should the company pay the ransom to retrieve the decryption key?

Date: Oct 6th, 2026	**Entry:** 2
Description	Analyzing a packet capture file
Tool(s) used	For this activity, I analyzed a packet capture file using Wireshark, a GUI-based network protocol analyzer. Wireshark is an essential cybersecurity tool because it enables analysts to monitor and evaluate network traffic, making it much easier to spot and investigate potential threats.
The 5 W's	Capture the 5 W's of an incident. • **Who** caused the incident? N/A • **What** happened? N/A • **When** did the incident occur? N/A • **Where** did the incident happen? N/A • **Why** did the incident happen? N/A

Additional notes	Using Wireshark for the first time was a rewarding experience. Although it presented a lot of information at first, I quickly grasped how to locate specific data based on what I was looking for. The process was quite repetitive and intuitive, I just needed to click on the main header or category related to my query. For instance, to examine protocol-specific data, I simply navigated to the Network or Internet layer section.

Date: Oct 6th, 2026	**Entry:** 3
Description	Capturing my first packet
Tool(s) used	For this activity, I utilized tcpdump to capture and inspect network traffic. Unlike GUI-based tools, tcpdump is a command-line network protocol analyzer. Much like Wireshark, it provides significant value in cybersecurity by allowing security analysts to efficiently capture, filter, and analyze live packet streams to investigate network behavior.
The 5 W's	Capture the 5 W's of an incident. • **Who** caused the incident? N/A • **What** happened? N/A • **When** did the incident occur? N/A • **Where** did the incident happen? N/A • **Why** did the incident happen? N/A
Additional notes	Using tcpdump can seem complicated when new commands are introduced. However, I quickly became familiar with the language tcpdump uses and immediately understood when to use sudo, knowing I lacked default

	permissions and needed it for the task. I also grasped options like -i. It was fun, and I feel that the way tcpdump syntax is structured is fairly easy for me to read, even if I don't understand everything yet. It's like learning a second language, even if you don't understand every part of the command, you can get an idea of what it is asking for just by reading it.

Date: Oct 8th, 2026	**Entry:** 4
Description	Investigate a suspicious file hash
Tool(s) used	In this activity, I utilized VirusTotal, a threat intelligence platform designed to analyze files and URLs for malicious artifacts like viruses, worms, and Trojans. VirusTotal allows analysts to rapidly verify whether an indicator of compromise (IoC), such as a file hash or domain, has already been flagged by the broader cybersecurity community. Working within the Detection and Analysis phase of the incident response lifecycle, I assumed the role of a Security Operations Center (SOC) analyst investigating a flagged file hash. Following an initial security alert, I leveraged VirusTotal to perform deeper analysis and determine whether the file posed a genuine threat.
The 5 W's	Capture the 5 W's of an incident. • **Who** caused the incident? An advanced threat actor Blacktech • **What** happened? A phishing email with a file hash incident • **When** did the incident occur? Thursday at 1:11 p.m. • **Where** did the incident happen? At a U.S financial services company

	• **Why** did the incident happen? The incident happened because an employee received an email containing an attachment. The attachment was a password-protected spreadsheet file. The spreadsheet's password was provided in the email. The employee downloaded the file, then entered the password to open the file. When the employee opened the file, a malicious payload was then executed on their computer.
Additional notes	1- How could this company prevent this type of incident? 2- Does this company offer updated phishing awareness training for employees?

Date: Oct 9, 2026	**Entry:** 5
Description	Investigating Phishing Alert ticket
Tool(s) used	I used VirusTotal as the primary threat intelligence tool to investigate a phishing alert ticket. By analyzing the file hash of the suspicious email attachment, VirusTotal enabled me to verify that the downloaded file was known malicious software.
The 5 W's	Capture the 5 W's of an incident. • **Who** caused the incident? Clyde West • **What** happened? A Phishing ticket alert • **When** did the incident occur? Wednesday at 09:30:14 AM • **Where** did the incident happen? At a U.S financial services company

	• **Why** did the incident happen? Previously, I received a phishing alert about a suspicious file being downloaded on an employee's computer. After investigating the email attachment file's hash, the attachment has already been verified to be malicious. Now that I have this information, you must follow your organization's process to complete your investigation and resolve the alert.
Additional notes	The Phishing alert is legit because I have verified it at VirusToal. The Phishing alert is legit because I reviewed the email details and it has a malicious file to download it.

Date: October 9th, 2026	**Entry:** 6
Description	Review a final report
Tool(s) used	N/A
The 5 W's	Capture the 5 W's of an incident. • **Who** caused the incident? An unknown individual • **What** happened? A data breach incident • **When** did the incident occur? on December 28, 2022, at 7:20 p.m. • **Where** did the incident happen? At a mid-sized retail company

	• **Why** did the incident happen? A vulnerability in the e-commerce web application allowed the attacker to perform a forced browsing attack and access customer transaction data by modifying the order number included in the URL string of a purchase confirmation page. This vulnerability allowed the attacker to access customer purchase confirmation pages, exposing customer data, which the attacker then collected and exfiltrated.
Additional notes	1) Should this company perform routine vulnerability scans and penetration testing? 2) Would be beneficial for this company to Implement access control mechanisms such as: - Implement allowlisting to allow access to a specified set of URLs and automatically block all requests outside of this URL range. - Ensure that only authenticated users are authorized access to content.

Date: Record the date of the journal entry.	**Entry:** Record the journal entry number.
Description	Provide a brief description about the journal entry.
Tool(s) used	List any cybersecurity tools that were used.

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Additional notes	Include any additional thoughts, questions, or findings.

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Reflections/Notes:

1. Were there any specific activities that were challenging for you? Why or why not?

Navigating tcpdump was the most challenging activity because learning and applying new command-line syntax all at once presents a steep learning curve. Working through complex flags and options required careful attention to detail. However, this hands-on experience laid a strong foundation, and I am confident that with continued practice and repetition, I will master command-line packet analysis.

2. Has your understanding of incident detection and response changed after taking this course?

My perspective on incident detection and response has evolved significantly. While I now appreciate the vast array of tools, protocols, and technical details required to respond effectively to threats, such as using VirusTotal to quickly cross reference suspicious file hashes, I was pleasantly surprised by the structured simplicity of incident logging. Coming from a different career field, seeing how concise and direct journal entries are was a major relief. As someone who naturally prefers clear, direct communication, this structured documentation style aligns perfectly with my strengths while making complex security investigations manageable.

3. Was there a specific tool or concept that you enjoyed the most? Why?

The concept I enjoyed the most was the incident journal and the sequential layout of its entries. Having a structured step by step timeline makes the entire investigation process clear and easy for anyone to follow. Following a playbook throughout the resolution process was also very straightforward. I really appreciate flowcharts because they present concepts that are easy to understand, even during high pressure situations. Ultimately, organizing complex technical details into a logical flow ensures that anyone reading the log can quickly understand what happened, what actions were taken, and how the incident was resolved.